



Health A to Z (Link: www.nhs.uk/health-a-to-z/)

NHS services (Link: www.nhs.uk/nhs-services/)

Healthy living (Link: www.nhs.uk/live-well/)

Low blood sugar (hypoglycaemia)

Low blood sugar (hypoglycaemia or a hypo) is usually where your blood sugar (glucose) is below 4mmol/L. It needs to be treated quickly to stop it getting worse, but you can usually treat it yourself.

You can also have blood sugar that's too high. This is called high blood sugar (hyperglycaemia) (Link: www.nhs.uk/conditions/high-blood-sugar-hyperglycaemia/).

Symptoms of low blood sugar

Low blood sugar usually affects people with diabetes (Link: www.nhs.uk/conditions/diabetes/) who take insulin or some other diabetes medicines. It's rare in people without diabetes.

Symptoms of low blood sugar can include:

- feeling hungry
- feeling dizzy
- feeling anxious or irritable
- sweating
- shaking
- tingling lips
- heart palpitations
- feeling tired or weak
- changes in your vision such as blurred vision
- feeling confused

If you have severe low blood sugar you may:

- have a seizure or fit
- become unconscious

Checking your blood sugar

If you have diabetes and get any symptoms of low blood sugar, check your blood sugar if you can. If it's below 4mmol/L you'll need to take steps to treat it.

How to treat low blood sugar yourself

If you have diabetes and get symptoms of low blood sugar (a hypo) or your blood sugar is below 4mmol/L:

1. Eat or drink something that will raise your blood sugar quickly, such as a small glass of fruit juice or sugary fizzy drink, 5 glucose or dextrose tablets, 4 large jelly babies, or 2 tubes of glucose gel.
2. Check your blood sugar after 10 to 15 minutes.
3. If your blood sugar is still below 4mmol/L, have another sugary drink or snack, and check again after 10 minutes.
4. Once your symptoms have improved and your blood sugar is above 4mmol/L, eat something that will keep your blood sugar up for longer, such as some biscuits, a sandwich, or your next meal if it's due.

How to treat severe low blood sugar

If someone has very low blood sugar (a severe hypo) and becomes unconscious:

1. Do not give them any food or drink as they will not be able to swallow safely.
2. Put them into the recovery position (Link: www.nhs.uk/tests-and-treatments/first-aid/recovery-position/).
3. Give them a glucagon injection straight away, if one is available and you know how to use it.
4. If they start to recover within 10 minutes of having a glucagon injection and can swallow safely, give them some food or drink that will raise their blood sugar.
5. Stay with them until they're fully recovered.

Call 999 if:

You think someone has very low blood sugar and they're not responding normally or they're unconscious, and:

- a glucagon injection is not available or you do not know how to use it
- you've given a glucagon injection and they have not recovered within 10 minutes
- they've been drinking alcohol

See a GP if:

- you have diabetes and you get low blood sugar (hypos) often, you've had severe hypos or you've had hypos at night
- you have diabetes and you do not always recognise the symptoms when you have low blood sugar
- you do not have diabetes and you get symptoms of low blood sugar such as shaking, sweating, a fast heartbeat and feeling confused

If you have diabetes, you can also speak to your diabetes care team for advice.

How to reduce your risk of low blood sugar

If you take medicine for diabetes, there are things you can do to reduce the risk of low blood sugar.

Do

- ✓ follow the treatment plan agreed with your diabetes care team, including adjusting your insulin dose when you need to
- ✓ check your blood sugar regularly
- ✓ carry something with you that will raise your blood sugar quickly, such as sugary drinks, sweets or glucose tablets
- ✓ carry a glucagon injection kit and medical ID with you
- ✓ make sure your family and friends know how to treat severe low blood sugar
- ✓ ask your care team about getting a continuous glucose monitor or flash monitor (Link: www.nhs.uk/tests-and-treatments/cgm-and-hcl-for-diabetes/) if you do not already have one
- ✓ if you have type 1 diabetes and often have problems with low blood sugar, ask your diabetes care team about switching to an insulin pump

Don't

- ✗ do not skip or delay meals
- ✗ do not drink too much alcohol – keep to the recommended guidelines of no more than 14 units of alcohol (Link: www.nhs.uk/live-well/alcohol-advice/calculating-alcohol-units/) a week, spread over 3 days or more
- ✗ do not drink alcohol without eating

If you drive, you'll need to check your blood sugar before each journey and then every 2 hours while driving. If you get low blood sugar, do not drive for at least 45 minutes

after your blood sugar is back to normal.

The DVLA has more information about diabetes and driving (Link: <https://www.gov.uk/government/publications/information-for-drivers-with-diabetes>).

How you get low blood sugar

If you have diabetes

Diabetes causes high blood sugar (hyperglycaemia) (Link: www.nhs.uk/conditions/high-blood-sugar-hyperglycaemia/). Diabetes medicines treat your condition by lowering your blood sugar, but they can sometimes make it go too low.

This is common if you take insulin (Link: www.nhs.uk/medicines/insulin/). It can also happen with some other diabetes medicines, such as gliclazide (Link: www.nhs.uk/medicines/gliclazide/) and glimepiride.

You're more likely to get low blood sugar if you:

- take too much insulin
- have problems with the way you inject insulin, such as injecting in the same place too often
- miss or delay meals, or do not eat enough carbohydrate
- do a lot of exercise without eating more carbohydrate or reducing your insulin dose
- drink a lot of alcohol, particularly without eating

Sometimes it's not clear why it happens.

If you do not have diabetes

It's rare to get low blood sugar if you do not have diabetes. It can sometimes be caused by conditions including:

- malnutrition (Link: www.nhs.uk/conditions/malnutrition/)
- Addison's disease (Link: www.nhs.uk/conditions/addisons-disease/)
- a growth (tumour) in your pancreas

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